VIGNAN'S INSTITUTE OF INFORMATION TECHNOLOGY (A)
Question Bank Mid-I Portion

Q. No	Questions	Marks	Section	Unit
1	Define Hilbert Space.	(2M)	**Section**-I	1
2	What is a Density Matrix?	(2M)	**Section**-I	1
3	Explain the significance of the Inner Product in Hilbert Space.	(2M)	**Section**-I	1
4	Differentiate between Pure State and Mixed State.	(2M)	**Section**-I	1
5	Explain the purpose of the Bloch Sphere in representing a qubit.	(2M)	**Section**-I	1
6	Give the clarity for the below question Illustrate the operation of the Hadamard Gate on computational basis states.	(2M)	**Section**-I	1
7	Explain the concept of Hilbert Space, basis vectors, orthogonality, completeness, and inner product with suitable mathematical expressions.	(6M)	**Section**-II	1
8	Describe the Density Matrix Formalism and derive the density matrix for a pure quantum state.	(6M)	**Section**-II	1
9	Explain the Bloch Sphere Representation of a qubit and derive the corresponding density matrix.	(6M)	**Section**-II	1
10	Derive the matrix representation and operation of Pauli-X, Pauli-Y, Pauli-Z, Hadamard, Phase (S), and T gates with suitable examples.	(6M)	**Section**-II	1
11	Explain the construction and working of the Controlled-NOT (CNOT) Gate with truth table, matrix representation, and quantum circuit.	(6M)	**Section**-II	1
12	Explain the working of the SWAP Gate with circuit diagram, truth table, and matrix representation.	(6M)	**Section**-II	1
13	Derive the operation of the Controlled-Z Gate and Toffoli Gate using suitable quantum circuits and matrix representations.	(6M)	**Section**-II	1
14	Analyze the generation of Bell States using Hadamard and CNOT gates. Justify the role of entanglement in Bell state preparation with suitable mathematical derivation and circuit diagram.	(6M)	**Section**-II	1
15	Define the No-Cloning Theorem.	(2M)	**Section**-I	2
16	What is Quantum Teleportation?	(2M)	**Section**-I	2
17	Explain why an unknown quantum state cannot be cloned.	(2M)	**Section**-I	2
18	Differentiate between classical copying and quantum cloning.	(2M)	**Section**-I	2
19	Explain the significance of Bell's Inequality.	(2M)	**Section**-I	2
20	Illustrate how entanglement is utilized in Quantum Teleportation.	(2M)	**Section**-I	2
21	Explain the No-Cloning Theorem and derive its mathematical	(6M)	**Section**-II	2

	proof using the inner product preservation property of unitary operators.			
22	Describe the cloning of basis states and superposition states. Explain why cloning fails for arbitrary quantum states with suitable mathematical analysis.	(6M)	**Section**-II	2
23	Explain the Quantum Teleportation protocol with the quantum circuit and describe the roles of Alice and Bob.	(6M)	**Section**-II	2
24	Derive the mathematical steps involved in the Quantum Teleportation algorithm and explain the state transformation after each quantum operation.	(6M)	**Section**-II	2
25	Explain Bell's Inequality, derive the CHSH inequality, and discuss its significance in quantum mechanics.	(6M)	**Section**-II	2
26	Derive the mathematical proof showing the violation of Bell's Inequality by quantum mechanics using an entangled Bell state.	(6M)	**Section**-II	2
27	Explain the implications of the No-Cloning Theorem and discuss its role in Quantum Communication and Quantum Cryptography.	(6M)	**Section**-II	2
28	Analyze the relationship among the No-Cloning Theorem, Quantum Teleportation, and Bell's Inequality. Justify how these principles establish the foundations of secure quantum communication.	(6M)	**Section**-II	2
29	Define the Deutsch Algorithm.	(2M)	**Section**-I	3
30	What is the objective of the Deutsch–Jozsa Algorithm?	(2M)	**Section**-I	3
31	Distinguish between Constant and Balanced Boolean Functions.	(2M)	**Section**-I	3
32	Explain the role of Phase Kickback in the Deutsch Algorithm.	(2M)	**Section**-I	3
33	Explain the Deutsch Algorithm with a neat quantum circuit and discuss the initialization of qubits, oracle operation, and final measurement.	(6M)	**Section**-II	3
34	Demonstrate the execution of the Deutsch–Jozsa Algorithm for a three-qubit input register by explaining the state transformations at every stage.	(6M)	**Section**-II	3
35	Explain the Deutsch–Jozsa Algorithm with a suitable quantum circuit and discuss each computational step in detail.	(6M)	**Section**-II	3
36	Derive the mathematical representation of the Deutsch Algorithm from the initial state through Hadamard transformation, oracle operation, and final measurement.	(6M)	**Section**-II	3